

Physiological Biomarkers for Personalized Vagus Nerve Stimulation in Neurorehabilitation

Addressing Feedback

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Specific Aims: Physiological Biomarkers for Personalized Vagus Nerve Stimulation in Neurorehabilitation Vagus nerve stimulation (VNS) has emerged as a promising neuromodulation strategy¹⁻³ for promoting neuroplasticity and functional recovery in individuals with neurological impairments such as stroke and spinal cord injury (SCI). While VNS-mediated rehabilitation has demonstrated lasting motor improvements in many patients^{2,4,5}, the field lacks objective clinical physiological biomarkers^{6,7} that can confirm vagal engagement and predict individual responsiveness to therapy. This limits the ability to optimize stimulation parameters and personalize treatment protocols. This project aims to address that gap by developing a multimodal platform capable of delivering precisely timed stimulation and recording neural and peripheral responses. Both implanted VNS^{3,8} and non-invasive transauricular nerve stimulation (tAN)⁹⁻¹⁵ will be evaluated. tAN is believed to stimulate the auricular branch of the vagus nerve, offering a non-surgical alternative to implanted VNS. Although tAN has gained popularity, the efficacy and mechanisms of tAN remain under debate due to anatomical variability and inconsistent evidence of central vagal engagement. By directly comparing tAN and implanted VNS, I aim to determine whether tAN reliably activates similar physiological pathways. This project addresses this gap through a three-aim strategy: first establishing a biomarker profile of implanted VNS engagement in clinical populations (Aim 1), then determining whether tAN produces overlapping physiological signatures in general population (Aim 2) and finally using these biomarkers to predict optimal stimulation parameters and rehabilitation outcomes prior to randomized controlled trials (Aim 3). Together, these aims will provide the first hypothesis-driven, biomarker framework for precision VNS neurorehabilitation. This research will focus on developing validated biomarkers of vagus nerve engagement through both central and peripheral modalities which include metrics such as electroencephalography (EEG)¹⁶⁻¹⁸, electromyography (EMG)¹⁹⁻²¹, electrocardiography (ECG)²²⁻²⁴, galvanic skin response (GSR), photoplethysmography (PPG), respiration, and pupillometry^{25,26}. These physiological signatures will be used to identify stimulation patterns associated with effective vagal engagement and therapeutic responsiveness, enabling individualized optimization of stimulation amplitude, frequency, pulse width, and timing.

Aim 1: Identify physiological biomarkers of implanted VNS engagement in chronic stroke and SCI populations. Develop and deploy the ReCorder recording and stimulation platform to measure stimulus-locked changes in EEG, EMG, ECG/HRV, GSR, PPG, respiration, and pupillometry during implanted VNS and sham stimulation using a fixed clinically relevant stimulation parameter set. Multimodal analyses will be used to determine whether stimulus-locked biomarkers can reliably distinguish active from sham stimulation across participants. **Hypothesis:** Implanted VNS will produce reproducible stimulus-locked physiological responses across autonomic, cortical, and neuromuscular modalities that significantly distinguish active stimulation from sham, establishing the existence of measurable multimodal biomarkers of vagal engagement.

Aim 2: Determine whether transauricular nerve stimulation (tAN) engages physiological pathways overlapping with implanted VNS. Apply the identical ReCorder acquisition and analysis framework from Aim 1 to general population receiving tAN and sham stimulations, then directly compare the temporal, spectral, and cross-modal biomarker features against implanted VNS signatures. Multimodal feature-space mapping will be used to identify which physiological components are shared versus distinct between stimulation modalities.

Hypothesis: Although tAN will engage subsets of autonomic and central pathways associated with implanted VNS, it will produce physiologically distinct multimodal signatures characterized by altered amplitudes, response latencies, and inter-modal coordination patterns, reflecting differences in neural recruitment and cutaneous stimulation mechanisms.

Aim 3: Determine whether acute biomarker responses can predict individualized therapeutic responsiveness and optimal VNS dosing. Collect multimodal physiological responses during a single pre-therapy stimulation parameter sweep session and use predictive modeling to relate acute biomarker features to subsequent rehabilitation outcomes. Analyses will identify biomarker profiles associated with therapeutic improvement and individualized parameter sensitivity. **Hypothesis:** Acute pre-therapy biomarker response profiles will predict post-rehabilitation outcomes and identify subject-specific optimal stimulation parameters with accuracy exceeding chance, with autonomic biomarkers including pupillometry and HRV serving as the strongest predictors of treatment responsiveness.

Completion of these aims will establish the first prospective biomarker framework for VNS neurorehabilitation and lay the foundation for the ReTune personalized stimulation platform for individualized parameter selection.

Research Strategy

A. SIGNIFICANCE

Stroke and Spinal Cord Injury

Approximately 300,000 individuals in the United States live with spinal cord injury (SCI), with nearly 18,000 new cases occurring annually^{27,28}. Over 50% of SCIs involve the cervical spinal cord, resulting in persistent motor and sensory impairments below the level of injury in most affected individuals²⁹. Stroke affects more than 7 million Americans, with approximately 650,000 new strokes occurring each year^{27,28}. Upper limb and gait impairments are common in both populations, affecting up to 80% of stroke survivors and a large proportion of individuals with cervical SCI^{30–32}. Although rehabilitation can restore meaningful function, many individuals experience persistent disability and reach a functional recovery plateau within months of injury, highlighting the need for more effective adjunct neurorehabilitation therapies^{33–36}.

VNS as a Rehabilitation Adjunct: Promise and Persistent Variability

Vagus nerve stimulation (VNS) has emerged as a powerful adjunct to motor rehabilitation, enhancing cortical plasticity and functional recovery in stroke and SCI^{1,2,4,37–42}. VNS promotes neuroplasticity through the timed release of acetylcholine, norepinephrine, and serotonin, which modulate synaptic strength in motor circuits when paired with correct movement attempts⁴³. Implanted VNS received FDA approval for upper limb stroke rehabilitation in 2021, and multiple trials have demonstrated meaningful gains in motor function^{2,3,5,44–46}. However, treatment outcomes vary widely, and a meaningful subset of patients fails to respond despite appropriate therapy³. Prior analyses suggest that conventional clinical factors such as lesion size, age, baseline impairment, and time since injury do not fully explain this variability or reliably predict therapeutic response^{42,47–49}. These findings suggest that variability in physiological responsiveness to VNS itself may play a major role in treatment efficacy. Individual differences in vagal anatomy, cortical network organization, and baseline excitability may alter each person's responsiveness to stimulation, rendering population-averaged parameters suboptimal for many individuals⁵⁰. The absence of objective physiological biomarkers of VNS engagement that can confirm whether stimulation is effectively engaging vagal and downstream autonomic or central pathways in a given individual, fundamentally limits the ability to personalize therapy and to identify patients who fail to show a physiological response to stimulation, non-responders, early in treatment. Consistent with this limitation, prior clinical work has explored empirical adjustments to stimulation amplitude to partially account for inter-subject variability, with Figure 1 illustrating an associated increase in observed motor recovery response rates under adjusted stimulation settings.

Critical Gap: No Ground-Truth Biomarker Framework for Implanted VNS in Rehabilitation Populations

Existing neurological injury biomarker studies have focused primarily on using single-modality measures electrocardiography (ECG) or electroencephalography (EEG) alone and none have characterized a comprehensive peripheral and autonomic biomarker profile of implanted VNS in rehabilitation cohorts^{6,7}. Without a validated ground-truth profile, it is impossible to know whether stimulation is engaging the vagus at a given parameter setting, let alone whether responses differ meaningfully across individuals. No validated multimodal biomarker framework currently exists for confirming vagal engagement during implanted VNS in

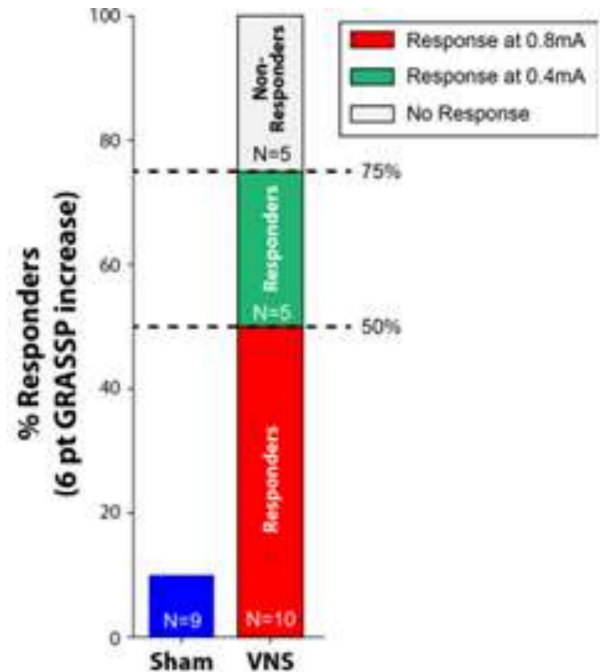


Fig 1. Lower VNS intensity enables clinical response in additional individual with spinal cord injury. Percent of participants achieving a clinically meaningful improvement (6-point GRASSP increase) during VNS-paired rehabilitation. Initial response rate at 0.8 mA was 50% (red); follow-on therapy at 0.4 mA converted 5 additional participants to responders (green), increasing overall efficacy to 75%.

stroke or SCI populations. Aim 1 directly addresses this gap by establishing the first systematic multimodal biomarker characterization of implanted VNS in chronic stroke and SCI, providing the ground truth that all subsequent aims depend upon.

The Non-Invasive Access Problem: Can tAN Serve as a Proxy?

Implanted VNS has demonstrated efficacy, but its surgical requirements limit broad accessibility.

Transauricular nerve stimulation (tAN) offers a non-invasive alternative by possibly targeting the auricular branch of the vagus nerve through the ear^{15,51}. Although tAN has gained clinical popularity, evidence for reliable central vagal engagement remains inconsistent. Additionally, no study to date has applied the same systematic multimodal approach to directly compare tAN and implanted VNS biomarker signatures⁹⁻¹³. The clinical relevance of this comparison is therefore open rather than established. If tAN were to reproduce key autonomic features of implanted VNS, it could support non-invasive physiological screening of vagal responsiveness prior to implantation. However, a more likely outcome is that tAN will exhibit a partially overlapping but fundamentally distinct physiological signature, reflecting differences in fiber recruitment, stimulation depth, and overall sensory arousal. In that case, the results would define the boundary conditions under which cutaneous auricular stimulation can or cannot serve as a surrogate for implanted VNS. Aim 2 directly tests these possibilities by comparing tAN and implanted VNS biomarker profiles using identical paradigms and analytical methods.

Critical Gap: No Validated Framework for Pre-Therapy Parameter Prediction

No validated framework currently exists for selecting VNS parameters based on an individual's physiological response profile prior to initiating therapy. Current clinical practice relies on population-average defaults or empirical titration during therapy, wasting valuable treatment time on suboptimal dosing and providing no early signal of non-response. None of the existing biomarker studies have prospectively linked pre-therapy acute physiological profiles to gait or motor outcomes in rehabilitation RCTs. The multimodal biomarker profiles validated in Aims 1 and 2 will directly fill this gap. Aim 3 will determine whether a single pre-therapy assessment session can forecast parameter-specific gait outcomes, enabling clinically actionable parameter recommendations before therapy begins.

B. INNOVATION

This proposal is organized around three independent aims that address distinct aspects of VNS physiology, cross-modal comparison, and predictive biomarker development.

1. First ground-truth multimodal biomarker characterization of implanted VNS in rehabilitation populations, supported by a purpose-built real-time platform. The ReRecorder platform developed to support Aim 1 will enable simultaneous recording of EEG, EMG, ECG, pupillometry, respiration, GSR, and PPG with sub-5 ms synchronization in stroke and spinal cord injury (SCI) populations. This will allow identification of coordinated autonomic, cortical, and neuromuscular response patterns that cannot be captured using single-modality approaches. The central goal is to determine whether implanted VNS produces a reproducible, system-wide physiological signature during rehabilitation-relevant stimulation.

2. First hypothesis-driven, direct comparison of tAN and implanted VNS using identical multimodal paradigms in a controlled general subject pool. No prior study has applied the same multimodal battery under the same recording conditions to directly compare tAN, implanted VNS, and sham. Using the Aim 1 implanted VNS profiles as ground truth, Aim 2 will resolve the ongoing debate about whether tAN elicits genuine central vagal activation and will precisely characterize which biomarker modalities tAN can and cannot reproduce.

3. Novel pre-therapy biomarker-based prediction of VNS parameters before gait rehabilitation. Aim 3 introduces a new paradigm: using a single pre-therapy assessment session to forecast which stimulation parameters will yield optimal gait outcomes for individuals before the RCT begins. This approach linking acute multimodal biomarker signatures to prospective therapy outcomes has not been attempted in rehabilitation VNS research and directly advances precision neuromodulation through the ReTune personalized dosing framework.

C. APPROACH

The three aims share a common experimental infrastructure: the ReRecorder Lab Streaming Layer software-synchronized multimodal recording platform, the ReStore implanted VNS system, and tAN/sham stimulation delivery across chronic stroke, chronic SCI, and general adult populations. Aim 1 establishes the ground-truth

biomarker profile using implanted VNS; Aim 2 tests whether tAN reproduces those profiles in a general comparison pool; Aim 3 applies the validated biomarker set to predict pre-therapy RCT outcomes. Shared methods participant recruitment, recording setup, artifact rejection, and data management are described under Aim 1 and referenced in subsequent aims.

Aim 1: Characterize Physiological Biomarkers of Vagal Engagement Using Implanted VNS in Clinical Populations

Objective and Rationale

Aim 1 has two interconnected objectives: (1a) develop and validate the ReRecorder multimodal recording and stimulation platform, and (1b) deploy it to characterize the physiological biomarker signatures of implanted VNS in chronic stroke and SCI populations. Reliable temporal alignment across signals and a validated stimulation delivery system are prerequisites for all biomarker analyses in Aims 2 and 3. Without a well-characterized implanted VNS biomarker profile, any comparison with tAN (Aim 2) or prediction of therapy outcomes (Aim 3) lacks a principled foundation.

ReRecorder Platform Development

The ReRecorder system will be developed in Python using Lab Streaming Layer (LSL) as the synchronization backbone, targeting end-to-end timing accuracy <5 ms. The platform integrates the following hardware streams: 8x8 high-density EMG (HD-EMG) arrays; 3-lead ECG; chest-belt respiration; finger-mounted GSR and PPG; and infrared video-based pupillometry (60 Hz), and BrainVision EEG. Stimulation delivery is controlled directly through ReRecorder, enabling automated parameter sweep execution (amplitude, frequency, and pulse width varied in randomized blocks every 6-10 s) with simultaneous event marker injection into all recording streams. A real-time quality-control dashboard monitors electrode impedance, per-modality trial counts, and signal integrity, enforcing pre-specified minimum thresholds before each session proceeds. Synchronization fidelity will be validated via hardware loopback testing before each recording session.

Study Design

Aim 1 uses a within-subject, repeated-measures design in which participants receiving implanted VNS undergo alternating blocks of active stimulation and sham stimulation. Stimulation is delivered in short, structured blocks separated by rest periods, with block order randomized across sessions. During each block, physiological signals are recorded continuously using the ReRecorder platform.

Recordings include a resting baseline period followed by stimulation epochs during which a fixed VNS parameter set (clinically used amplitude, frequency, and pulse width) is applied. Sham blocks use identical timing without active stimulation delivery. This structure enables direct comparison of stimulus-locked physiological responses within the same individual.

All signals (EEG, EMG, ECG/HRV, GSR, PPG, respiration, and pupillometry) are recorded continuously across conditions with synchronized acquisition. Primary analyses focus on identifying time-locked changes relative to stimulation onset and comparing active versus sham response profiles within subjects.

Participants

Two deployment pipelines will be used: (1) implanted VNS in chronic SCI participants (SCI-AFOS); (2) implanted VNS in chronic stroke participants (Stroke-AFOS). SCI cohort: up to 19 chronic participants (ages 21-64); stroke cohort: up to 19 (ages 26-70). All participants provide informed consent and undergo safety screening prior to each session. Exclusion criteria include stimulation contraindications and significant neurological or psychiatric comorbidities beyond the qualifying diagnosis.

Data Analysis

Biomarker features will be extracted per modality as follows. 64-Channel EEG recorded throughout stimulation paradigm. EMG recorded from the laryngeal region. ECG/HRV: RMSSD, SDNN, and LF/HF ratio computed from stimulus-locked windows. Pupillometry: time-locked dilation or constriction amplitude and latency. GSR/PPG: Phasic SCR amplitude and PPG pulse amplitude, baseline-corrected to sham condition. Respiration: rate and depth changes with cycle-based phase segmentation. Acute effects will be evaluated with proper statistics comparing active and sham conditions across parameter levels. Longitudinal changes will be assessed with linear mixed-effects models. All analyses will include baseline-corrected sham conditions to control for non-specific arousal effects.

Expected Results and Pitfalls

Implanted VNS is expected to produce reliable stimulus-locked changes across multiple physiological modalities. These responses are anticipated to scale systematically with stimulation parameters, reflecting dose-dependent effects of VNS on autonomic, cortical, and neuromuscular systems. Among the measured signals, pupillometry and heart rate variability (HRV) are expected to show the most consistent and robust responses across participants. Electromyography (EMG) is expected to provide additional insight into peripheral neuromuscular modulation during stimulation.

Several potential challenges are anticipated. First, ensuring precise temporal alignment across recording systems is critical for detecting stimulus-locked effects. This will be addressed using hardware synchronization checks, including loopback validation and external trigger signals. External trigger signals refer to transistor–transistor logic (TTL) pulses, which provide a hardware-based timing reference used to synchronize stimulation onset with physiological recordings across devices.

Second, physiological recordings may be affected by motion or electrical artifacts. These will be mitigated using independent component analysis (ICA) and by enforcing minimum trial counts to ensure stable feature estimation.

Finally, recruitment and regulatory approval timelines for implanted VNS participants may introduce delays. To mitigate this risk, platform validation and initial analyses will begin with the healthy adult tAN cohort, which is already approved for data collection, while clinical approvals for implanted cohorts are finalized.

Aim 2: Validate Non-Invasive Transauricular Nerve Stimulation Biomarker Signatures Against the Implanted VNS Ground Truth in a Healthy Subject Comparison Pool

Objective and Rationale

Aim 2 will determine whether tAN produces biomarker signatures that are consistent with implanted VNS responses established in Aim 1 and will precisely characterize which modalities tAN can and cannot reproduce. A general population will be used to isolate the effect of stimulation modality from the confounds of neurological injury, providing the cleanest possible test of whether tAN engages the same physiological pathways as implanted VNS. The question is not simply whether tAN produces any response, but whether its response profile is sufficiently consistent with implanted VNS to serve as a non-invasive proxy for vagal engagement assessment.

Study Design

Aim 2 applies the same ReCorder recording framework used in Aim 1 to a cohort of general population adults. Participants receive transauricular nerve stimulation (tAN) and sham stimulation while undergoing identical recording and task conditions, including resting state and possible auditory cue–based paradigms. All physiological signals are acquired using the same hardware and synchronization methods as in Aim 1.

Stimulation conditions include three interleaved and randomized blocks: (1) tAN delivered via surface electrodes placed on the left ear (targeting the auricular region associated with vagal afferents), (2) sham stimulation delivered at a non-vagal site using a matched electrode configuration, and (3) a sensory control condition delivered on the finger to control for generalized cutaneous stimulation effects. Both active and control conditions are intensity-matched to produce comparable sensory perception across sites.

Block order is randomized within and across participants to minimize order and habituation effects. Stimulation and recording epochs are interleaved with rest periods to allow physiological signals to return to baseline between trials.

Biomarker profiles derived from tAN and sham sessions will be directly compared against the implanted VNS reference profiles generated in Aim 1 using a shared analytic pipeline. This enables quantitative assessment of similarities and differences in temporal dynamics, amplitude, and cross-modal coordination across stimulation modalities.

Participants

Up to 50 general population adults (UTD IRB #25-31, approved) will complete tAN active and sham sessions in a within-subject design. Exclusion criteria include vagal stimulation contraindications, significant cardiovascular or autonomic disorders, and psychiatric diagnoses that could confound autonomic or pupillary measures.

Data Analysis

The same feature extraction pipeline as Aim 1 will be applied to all Aim 2 recordings. Acute tAN effects will be evaluated with repeated-measures models comparing active and sham conditions across recording metrics. A cross-modality comparison framework will then assess the degree of overlap between tAN and implanted VNS biomarker profiles: effect sizes (Cohen's d) for each biomarker feature will be compared across stimulation modalities, and Bayesian equivalence testing will quantify the degree to which tAN responses are interchangeable with implanted VNS responses at each modality. Machine learning models (random forest) will identify which biomarker features best differentiate tAN from implanted VNS and which are shared, producing a ranked divergence profile across modalities. The resulting classification of 'shared' versus 'modality-specific' biomarkers will directly inform which features should be prioritized in the Aim 3 predictive framework.

Expected Results and Pitfalls

tAN is expected to reproduce autonomic biomarker signatures as shown in Fig 2. As well as (HRV, GSR, PPG, Pupil) observed with implanted VNS, reflecting shared activation of peripheral vagal afferents. Neuromuscular biomarkers (EMG amplitude) are expected to diverge, as these require central vagal engagement that tAN may not reliably produce. This pattern would support tAN as a non-invasive autonomic-level proxy while clarifying its limitations relative to implanted VNS. If tAN biomarkers converge broadly with implanted VNS across all modalities, this would provide strong evidence for tAN as a complete non-invasive substitute and strengthen its clinical candidacy. Recruitment of the general population cohort is already IRB-approved and underway, mitigating the risk of delays. If tAN responses are highly variable across individuals, subgroup analyses stratified by perception analysis and general parameter levels will be investigated for correlations.

Aim 3: Predict Optimal VNS Stimulation Parameters and Rehabilitation Outcomes Using Pre-Therapy Acute Biomarker Profiles

Objective and Rationale

Aim 3 will determine whether a single pre-therapy acute biomarker assessment session can predict which VNS stimulation parameters will yield optimal gait outcomes for participants about to begin a VNS-paired gait RCT. Building directly on the candidate biomarkers validated in Aims 1 and 2 such as pupil response, HRV indices, EMG responses, GSR phasic amplitude, and PPG. This aim will construct individual biomarker response profiles from pre-therapy parameter sweeps and test whether those profiles prospectively forecast each participant's treatment response and optimal parameter settings. Success would enable clinicians to personalize stimulation parameters before therapy begins, eliminating trial-and-error and reducing time on suboptimal dosing.

Study Design

Participants enrolled in the SCI-AFOS or Stroke-AFOS VNS or paired gait RCT will complete a single pre-therapy biomarker assessment session (1-2 hours) immediately before their first therapy visit. Stimulation parameter sweeps varying amplitude (0.2-1.2 mA), frequency (10-30 Hz), and pulse width (100-250 μ s) in randomized blocks will be delivered while the full ReRecorder multimodal recording suite is active. Acute physiological responses to each parameter combination will be extracted to form a high-dimensional biomarker response profile per participant. Following RCT completion (12-16 weeks), gait outcome measures (10-Meter Walk Test [10MWT], 6-Minute Walk Test [6MWT], Timed Up and Go [TUG]) will serve as clinical outcomes for model training and evaluation.

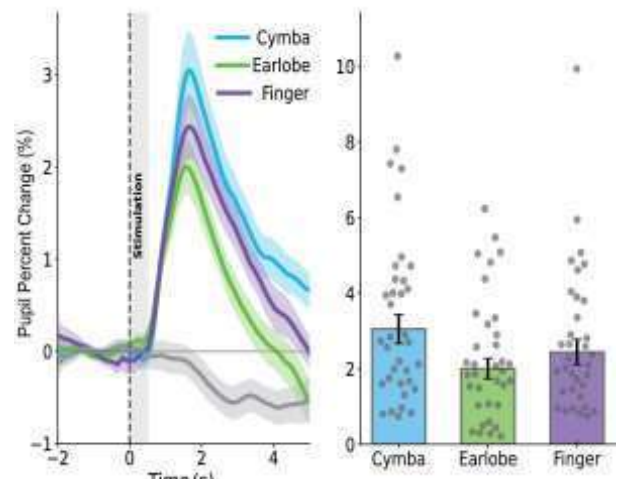


Fig 2. Pupil dilation response to cutaneous stimulation across electrode placement sites. (Left) Mean pupil percent change ($\pm$ SEM) time-locked to stimulation onset (dashed line) for cymba conchae (blue), earlobe (green), and finger (purple) placements. Gray shading indicates the baseline no stimulation condition. (Right) Peak pupil dilation amplitude per condition with individual participant data overlaid. No significant differences were found when comparing the three stimulation conditions.

Participants

All participants enrolled in SCI-AFOS, Stroke-AFOS, or gait RCT who complete the pre-therapy assessment will be eligible. A minimum of 5 participants per cohort (SCI, stroke) is targeted based on minimum power analyses for regression models with 10-15 biomarker features. Participants who withdrew from the RCT before the mid-therapy timepoint will be excluded from outcome prediction analyses but included in pre-therapy biomarker characterization analyses.

Data Analysis

Feature extraction will quantify per participant and parameter combination: EEG, pupil response peak amplitude and latency, ECG RMSSD and LF/HF ratio, EMG amplitude, GSR peak amplitude, and PPG pulse amplitude change. Parameter-level features will be summarized into a per-participant profile vector. Predictive models (multivariate linear regression, LASSO, random forest) will be trained to map pre-therapy biomarker profiles to post-RCT gait outcome scores. Leave-one-out cross-validation will assess generalizability, and permutation-based feature importance will identify which biomarker modalities best predict parameter-specific treatment response. Exploratory clustering will test whether distinct biomarker subtypes (e.g., strong autonomic responders vs. strong neuromuscular responders) correspond to differential therapy outcomes, supporting a subgroup-level parameter recommendation framework.

Expected Results and Pitfalls

Pre-therapy biomarker profiles are expected to predict post-RCT gait outcomes with accuracy exceeding chance, with pupillometry and HRV anticipated to be the most sensitive predictors of vagal engagement and therapy response. Participants with stronger acute responses to specific parameter combinations are expected to benefit most from those parameters during therapy. Validated models will provide a biomarker-guided parameter selection framework directly applicable to future RCTs and to clinical practice through the ReTune platform. Potential pitfalls: limited sample size from a single RCT may constrain model generalizability, mitigated by regularization (LASSO/ridge) and emphasis on cross-validated effect sizes. If individual modalities are insufficient as predictors, composite multimodal scores will be developed via PCA or sparse CCA. If full parameter sweep sessions are not tolerated, a shorter passive resting-state protocol will serve as a fallback, leveraging spontaneous autonomic variability as the predictor feature set.

Summary

Aim 3 will establish whether pre-therapy acute biomarker profiles can serve as personalized predictors of optimal VNS parameters, directly linking the biomarker science of Aims 1 and 2 to clinical decision-making before gait rehabilitation begins. Successful completion will provide the first prospective evidence that physiological phenotyping prior to an RCT can guide stimulation parameter selection a critical step toward precision neuromodulation and the development of the ReTune personalized dosing platform.

Shared Pitfalls Across All Aims

Regulatory delays for implanted VNS studies will be managed by initiating tAN only and healthy cohort data collection (both already IRB-approved) while clinical approvals are pending. Participant dropout across multi-session protocols will be mitigated through flexible scheduling and compensation, with interim attrition analyses at prespecified checkpoints. Data quality failures (electrode impedance drift, signal dropout) will be flagged by automated quality-control routines in ReCorder, with pre-specified minimum trial counts enforced per modality. All physiological analyses will include baseline-corrected sham conditions to control for non-specific arousal effects.

Conclusion

Successful completion of these aims will establish a cohesive framework for quantifying vagus nerve stimulation across implanted and non-invasive modalities and link physiological engagement to therapeutic outcomes. Aim 1 will define the physiological signature of implanted VNS in rehabilitation populations. Aim 2 will determine the extent to which tAN stimulation reproduces or diverges from this signature, establishing the boundaries of non-invasive vagal engagement. Aim 3 will translate these biomarkers into a predictive framework for optimal stimulation parameters and anticipating rehabilitation outcomes prior to therapy initiation. Together, these aims will move the field from empirically tuned stimulation toward objective, physiology-driven neuromodulation, enabling systematic measurement, comparison, and personalization of VNS-based therapies.

Citations:

1. Engineer, N. D. *et al.* Reversing pathological neural activity using targeted plasticity. *Nature* **470**, 101–104 (2011).
2. Dawson, J. *et al.* Safety, feasibility, and efficacy of vagus nerve stimulation paired with upper-limb rehabilitation after ischemic stroke. *Stroke* **47**, 143–150 (2016).
3. Kilgard, M. P. *et al.* Closed-loop vagus nerve stimulation aids recovery from spinal cord injury. *Nature* 1–7 (2025) doi:10.1038/s41586-025-09028-5.
4. Dawson, J. *et al.* Vagus nerve stimulation paired with rehabilitation for upper limb motor function after ischaemic stroke (VNS-REHAB): a randomised, blinded, pivotal, device trial. *The Lancet* **397**, 1545–1553 (2021).
5. Dawson, J. *et al.* Vagus nerve stimulation paired with rehabilitation for upper limb motor impairment and function after chronic ischemic stroke: subgroup analysis of the randomized, blinded, pivotal, VNS-REHAB device trial. *Neurorehabil. Neural Repair* **37**, 367–373 (2023).
6. Hammond, E. J., Uthman, B. M., Reid, S. A. & Wilder, B. J. Electrophysiological Studies of Cervical Vagus Nerve Stimulation in Humans: I. EEG Effects. *Epilepsia* **33**, 1013–1020 (1992).
7. Ravan, M., Sabesan, S. & D'Cruz, O. On Quantitative Biomarkers of VNS Therapy Using EEG and ECG Signals. *IEEE Trans. Biomed. Eng.* **64**, 419–428 (2017).
8. Sivaji, V. *et al.* ReStore: A wireless peripheral nerve stimulation system. *J. Neurosci. Methods* **320**, 26–36 (2019).
9. Yap, J. Y. Y. *et al.* Critical Review of Transcutaneous Vagus Nerve Stimulation: Challenges for Translation to Clinical Practice. *Front. Neurosci.* **14**, 284 (2020).
10. Badran, B. W. & Austelle, C. W. The Future Is Noninvasive: A Brief Review of the Evolution and Clinical Utility of Vagus Nerve Stimulation. *Focus J. Life Long Learn. Psychiatry* **20**, 3–7 (2022).
11. Gurtubay, I. G. *et al.* Immediate effects and duration of a short and single application of transcutaneous auricular vagus nerve stimulation on P300 event related potential. *Front. Neurosci.* **17**, (2023).

12. Farmer, A. D. *et al.* International Consensus Based Review and Recommendations for Minimum Reporting Standards in Research on Transcutaneous Vagus Nerve Stimulation (Version 2020). *Front. Hum. Neurosci.* **14**, 568051 (2020).
13. Baig, S. S. *et al.* tVNS in Stroke: A Narrative Review on the Current State and the Future. *Stroke* **54**, 2676–2687 (2023).
14. He, W. *et al.* Auricular Acupuncture and Vagal Regulation. *Evid.-Based Complement. Altern. Med. ECAM* **2012**, 786839 (2012).
15. Peuker, E. T. & Filler, T. J. The nerve supply of the human auricle. *Clin. Anat. N. Y. N* **15**, 35–37 (2002).
16. Vanneste, S., Martin, J., Rennaker, R. L. & Kilgard, M. P. Pairing sound with vagus nerve stimulation modulates cortical synchrony and phase coherence in tinnitus: An exploratory retrospective study. *Sci. Rep.* **7**, 17345 (2017).
17. Morrison, E. G. *et al.* Characterization of Vagus Nerve Stimulation (VNS) Dose-Dependent Effects on EEG Power Spectrum and Synchronization. *Biomedicines* **12**, 557 (2024).
18. Bartolomei, F. *et al.* How does vagal nerve stimulation (VNS) change EEG brain functional connectivity? *Epilepsy Res.* **126**, 141–146 (2016).
19. Guerrero-Mendez, C. D. & Ruiz-Olaya, A. F. Coherence-based connectivity analysis of EEG and EMG signals during reach-to-grasp movement involving two weights. *Brain-Comput. Interfaces* **9**, 140–154 (2022).
20. Konakoğlu, G., Özden, A. V., Solmaz, H. & Bildik, C. The effect of auricular vagus nerve stimulation on electroencephalography and electromyography measurements in healthy persons. *Front. Physiol.* **14**, (2023).
21. Udina, E., D’Amico, J., Bergquist, A. J. & Gorassini, M. A. Amphetamine increases persistent inward currents in human motoneurons estimated from paired motor-unit activity. *J. Neurophysiol.* **103**, 1295–1303 (2010).

22. Verrier, R. L., Nearing, B. D., Olin, B., Boon, P. & Schachter, S. C. Baseline elevation and reduction in cardiac electrical instability assessed by quantitative T-wave alternans in patients with drug-resistant epilepsy treated with vagus nerve stimulation in the AspireSR E-36 trial. *Epilepsy Behav.* **62**, 85–89 (2016).
23. Agarwal, R., Mokelke, E., Ruble, S. B. & Stolen, C. M. Vagal Nerve Stimulation Evoked Heart Rate Changes and Protection from Cardiac Remodeling. *J Cardiovasc. Transl. Res.* **9**, 67–76 (2016).
24. Genç, A., Uçan Tokuç, F. E. & Korucuk, M. Effects of vagal nerve stimulation parameters on heart rate variability in epilepsy patients. *Front. Neurol.* **15**, 1490887 (2024).
25. Sharon, O., Fahoum, F. & Nir, Y. Transcutaneous Vagus Nerve Stimulation in Humans Induces Pupil Dilation and Attenuates Alpha Oscillations. *J. Neurosci. Off. J. Soc. Neurosci.* **41**, 320–330 (2021).
26. Lloyd, B., Wurm, F., De Kleijn, R. & Nieuwenhuis, S. Short-term transcutaneous vagus nerve stimulation increases pupil size but does not affect EEG alpha power: A replication of Sharon et al. (2021, Journal of Neuroscience). *Brain Stimulat.* **16**, 1001–1008 (2023).
27. Tsao, C. W. *et al.* Heart disease and stroke statistics—2023 update: a report from the American Heart Association. *Circulation* **147**, e93–e621 (2023).
28. National Spinal Cord Injury Statistical Center. *Traumatic Spinal Cord Injury Facts and Figures at a Glance.* (2022).
29. Alizadeh, A., Dyck, S. M. & Karimi-Abdolrezaee, S. Traumatic spinal cord injury: an overview of pathophysiology, models and acute injury mechanisms. *Front. Neurol.* **10**, 282 (2019).
30. Ingram, L. A., Butler, A. A., Brodie, M. A., Lord, S. R. & Gandevia, S. C. Quantifying upper limb motor impairment in chronic stroke: a physiological profiling approach. *J. Appl. Physiol.* **131**, 949–965 (2021).
31. Chohan, S. A., Venkatesh, P. K. & How, C. H. Long-term complications of stroke and secondary prevention: an overview for primary care physicians. *Singapore Med. J.* **60**, 616 (2019).
32. Alizadeh, A., Dyck, S. M. & Karimi-Abdolrezaee, S. Traumatic spinal cord injury: an overview of pathophysiology, models and acute injury mechanisms. *Front. Neurol.* **10**, 282 (2019).
33. Stockley, R., Peel, R., Jarvis, K. & Connell, L. Current therapy for the upper limb after stroke: a cross-sectional survey of UK therapists. *BMJ Open* **9**, e030262 (2019).

34. Nas, K., Yazmalar, L., Şah, V., Aydın, A. & Öneş, K. Rehabilitation of spinal cord injuries. *World J. Orthop.* **6**, 8 (2015).
35. Langhorne, P., Bernhardt, J. & Kwakkel, G. Stroke rehabilitation. *The Lancet* **377**, 1693–1702 (2011).
36. Thuret, S., Moon, L. D. & Gage, F. H. Therapeutic interventions after spinal cord injury. *Nat. Rev. Neurosci.* **7**, 628–643 (2006).
37. Epperson, J. D. *et al.* Characterization of an algorithm for autonomous, closed-loop neuromodulation during motor rehabilitation. *Neurorehabil. Neural Repair* **38**, 493–505 (2024).
38. Epperson, J. *et al.* Characterization of an algorithm for autonomous, closed-loop neuromodulation during stroke rehabilitation. *Arch. Phys. Med. Rehabil.* **102**, e98 (2021).
39. Porter, B. A. *et al.* Repeatedly pairing vagus nerve stimulation with a movement reorganizes primary motor cortex. *Cereb. Cortex* **22**, 2365–2374 (2012).
40. De Ridder, D., Vanneste, S., Engineer, N. D. & Kilgard, M. P. Safety and efficacy of vagus nerve stimulation paired with tones for the treatment of tinnitus: a case series. *Neuromodulation J. Int. Neuromodulation Soc.* **17**, 170–179 (2014).
41. Hariz, M. I. Complications of deep brain stimulation surgery. *Mov. Disord. Off. J. Mov. Disord. Soc.* **17**, S162–S166 (2002).
42. Hays, S. A., Rennaker II, R. L. & Kilgard, M. P. How to fail with paired VNS therapy. *Brain Stimulat.* **16**, 1252–1258 (2023).
43. Schambra, H. M. & Hays, S. A. Vagus nerve stimulation for stroke rehabilitation: Neural substrates, neuromodulatory effects and therapeutic implications. *J. Physiol.* **603**, 723–735 (2025).
44. Baylor Research Institute. *Wireless Nerve Stimulation Device To Enhance Recovery After Stroke*. <https://clinicaltrials.gov/study/NCT04534556> (2023).
45. Baylor Research Institute. *Targeted Plasticity Therapy for Upper Limb Rehabilitation in Spinal Cord Injuries*. <https://clinicaltrials.gov/study/NCT04288245> (2023).

46. Dawson, J. *et al.* Vagus nerve stimulation paired with rehabilitation for upper limb motor function after ischaemic stroke (VNS-REHAB): a randomised, blinded, pivotal, device trial. *The Lancet* **397**, 1545–1553 (2021).
47. Meyers, E. C. *et al.* Vagus nerve stimulation enhances stable plasticity and generalization of stroke recovery. *Stroke* **49**, 710–717 (2018).
48. Dickie, D. A. *et al.* An exploratory study of predictors of response to vagus nerve stimulation paired with upper-limb rehabilitation after ischemic stroke. *Sci. Rep.* **9**, 15902 (2019).
49. Engineer, N. D. *et al.* Targeted vagus nerve stimulation for rehabilitation after stroke. *Front. Neurosci.* **13**, 280 (2019).
50. Patros, M., Sivathamboo, S., Simpson, H. D., O'Brien, T. J. & Macefield, V. G. The physiology, anatomy and stimulation of the vagus nerve in epilepsy. *J. Physiol.* **603**, 2201–2217 (2025).
51. He, W. *et al.* Auricular acupuncture and vagal regulation. *Evidence-Based Complement. Altern. Med.* **2012**, 786839 (2012).